Electronic Circuits and Applications

Syllabus

SY ECA Syllabus

Why are electronic circuits and amplifiers crucial in engineering?

Here are a few key reasons:

Communication Systems: Amplifiers are fundamental components in nearly all communication systems. They are used to boost weak signals received from antennas or to strengthen signals before transmission over long distances, ensuring clear and reliable communication. Without amplifiers, phenomena like long-distance radio broadcasts, mobile phone calls, and internet data transmission would not be possible.

Amplifiers

Control Systems: In control systems, electronic circuits and amplifiers are vital for processing sensor signals and driving actuators. For instance, an amplifier can take a small error signal from a sensor (e.g., temperature, pressure, position) and amplify it to a level sufficient to control a motor, valve, or heating element, thereby maintaining a desired system state.

Amplifiers

Consumer Electronics: From the smartphone in your pocket to the television in your living room, consumer electronics are replete with electronic circuits and amplifiers. Amplifiers are used in audio systems to power speakers, in display technologies to drive pixels, and in countless other applications to process and condition signals for various functions.

Amplifiers

Medical Devices: Electronic circuits are at the heart of many medical devices, from diagnostic equipment like ECG and MRI machines to therapeutic devices like pacemakers and defibrillators. Amplifiers in these devices are critical for detecting and processing faint biological signals, enabling accurate diagnosis and effective treatment.

Amplifiers

Instrumentation and Measurement: In scientific and industrial instrumentation, electronic circuits are essential for amplifying, filtering, and converting signals from sensors into measurable quantities. This allows engineers and researchers to accurately measure physical parameters such as voltage, current, temperature, and strain.

Amplifiers

Power Electronics: While distinct, power electronics heavily rely on the principles of electronic circuits and the use of power amplifiers to efficiently convert and control electrical power. This is crucial for applications like renewable energy systems (solar inverters, wind turbine converters), electric vehicles, and industrial motor drives.

Need for Biasing in BJT

- What is a BJT?
- NPN and PNP structures (very brief)
- Terminals: Emitter, Base, Collector
- Transistor as an Amplifier

Understanding BJT Basics

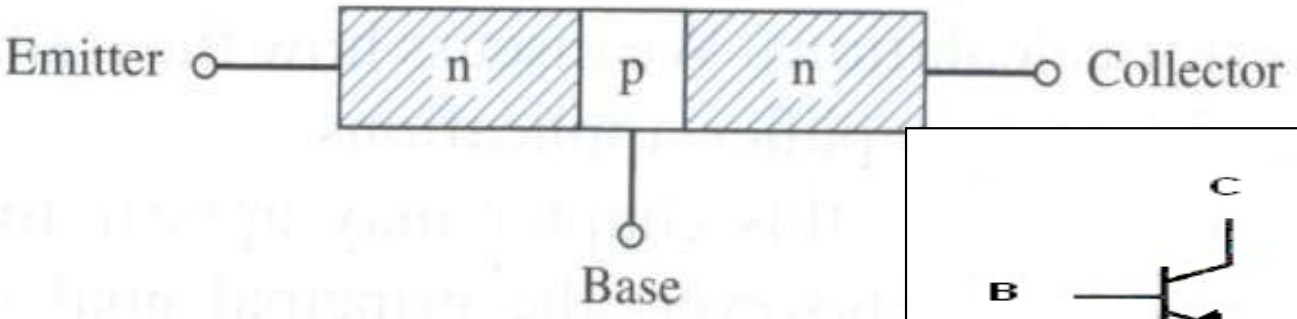
- **Bipolar Junction Transistors (BJTs):** Three-terminal semiconductor devices used for amplification or switching.

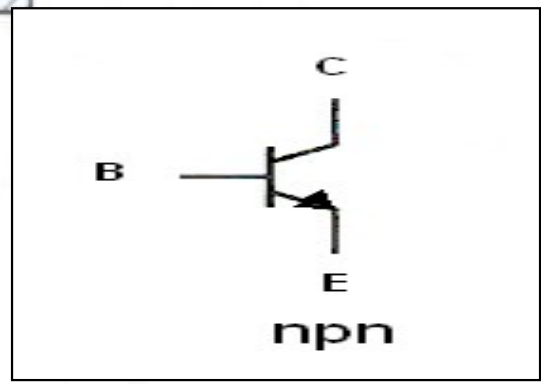
NPN & PNP

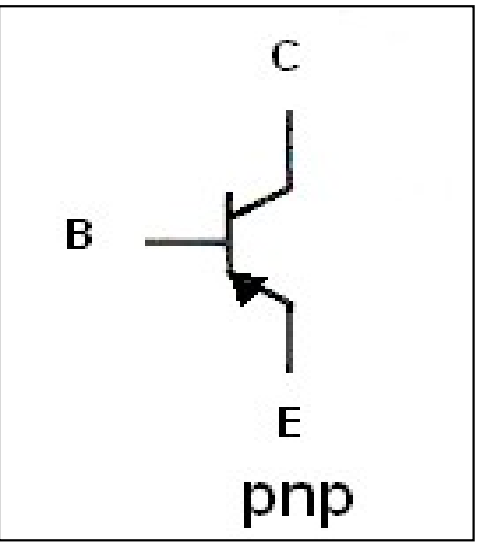
- **Why they are important:** Form the backbone of many analog electronic circuits.

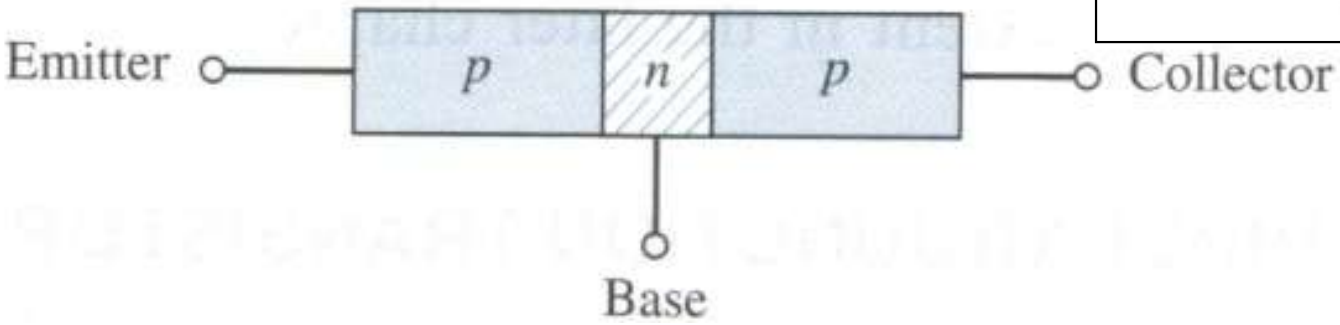
Transistor Structures

- The **bipolar junction transistor (BJT)** has three separately doped regions and contains two pn junctions.
- Bipolar transistor is a 3-terminal device.
  - Emitter (E)
  - Base (B)
  - Collector (C)









3 Regions of Operation

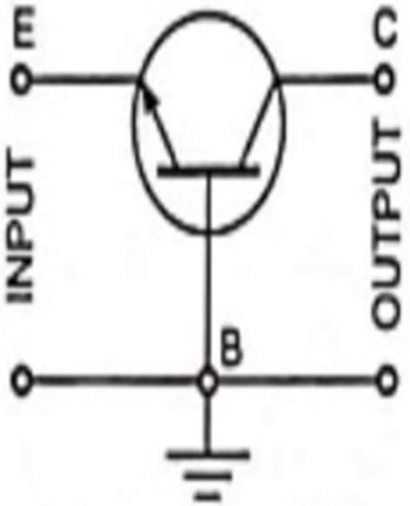
- **Active**
  Operating range of the amplifier.
  Base-Emitter Junction forward biased.
  Collector-Base Junction reverse biased

- **Cutoff**
  The amplifier is basically off. There is voltage but little current.
  Both junctions reverse biased

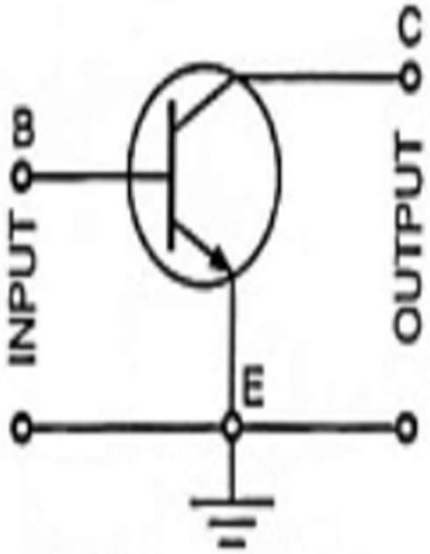
- **Saturation**
  The amplifier is full on. There is little voltage but lots of current.
  Both junctions forward biased

3 Configurations

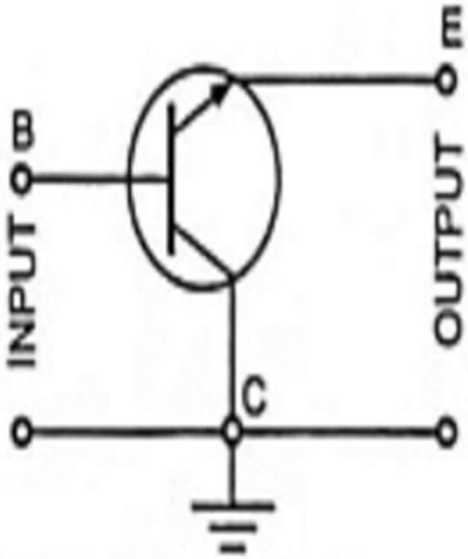
- **Common Emitter CE**
- **Common Base CB**
- **Common Collector CC**



(a) Common Base Configuration



(b) Common Emitter Configuration

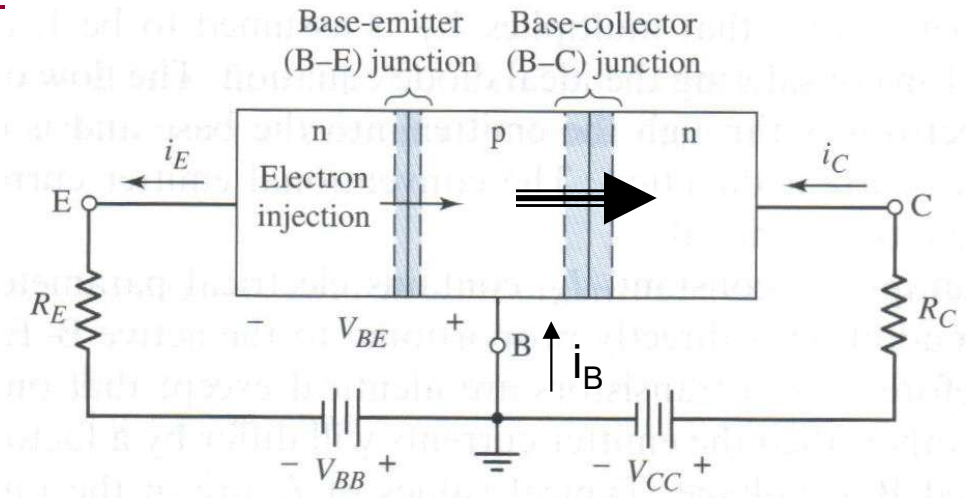


(c) Common Collector Configuration

OPERATIONS - npn

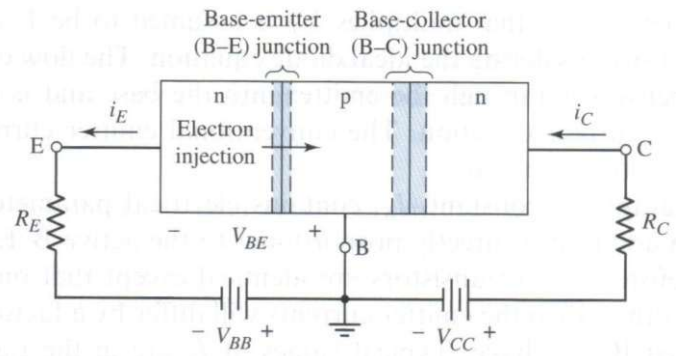
ACTIVE MODE

- The **base-emitter (B-E) junction** is forward biased and the **base-collector (C-B) junction** is reverse-biased,.
- Since the B-E junction is forward biased, electrons from the emitter are injected across the B-E junction into the base → I_E
- Once in the base region, the electrons are quickly accelerated through the base due to the reverse-biased C-B region → I_C



- Some electrons, in passing through the base region, recombine with majority carrier holes in the base. This produces the current → I_B

MATHEMATICAL EXPRESSIONS



$$I_E = I_S [e^{V_{BE} / V_T} - 1] = I_S e^{V_{BE} / V_T}$$

Based on KCL: $I_E = I_C + I_B$

No. of electrons crossing the base region and then directly into the collector region is a constant factor β of the no. of electrons exiting the base region

$$I_C = \beta I_B$$

No. of electrons reaching the collector region is directly proportional to the no. of electrons injected or crossing the base region.

$$I_C = \alpha I_E$$

Ideally $\alpha = 1$, but in reality it is between 0.9 and 0.998.

Based on KCL: $I_E = I_C + I_B$ $I_C = \beta I_B$ $I_C = \alpha I_E$

$I_E = \beta I_B + I_B = I_B(\beta + 1)$ ⟹ $I_E = I_B(\beta + 1)$

Now

With $I_C = \beta I_B \rightarrow I_B = I_C / \beta$

Hence,

$I_E = [\, I_C / \beta \,] \; (\beta + 1)$

$I_C = I_E [\, \beta \; / \; \beta + 1 \,]$

Comparing with $I_C = \alpha I_E$ ⟹ $\alpha = [\, \beta \; / \; \beta + 1 \,]$

EXAMPLE

Calculate the collector and emitter currents, given the base current and current gain.
Assume a common-base current gain $\alpha = 0.97$ and a base current of $i_B = 25\ \mu A$.
Also assume that the transistor is biased forward in the forward active mode.

$I_C = \alpha I_E$ $\qquad$ $I_C = \beta I_B$

Solution: The common-emitter current gain is $\beta = \frac{\alpha}{1-\alpha} = \frac{0.97}{1-0.97} = 32.33$

The collector current is $i_C = \beta i_B = 32.33 \times 25 = 808.25\ \mu A$

And the emitter current is $i_E = i_B + i_C = 25 + 808.25 = 833.25\ \mu A$

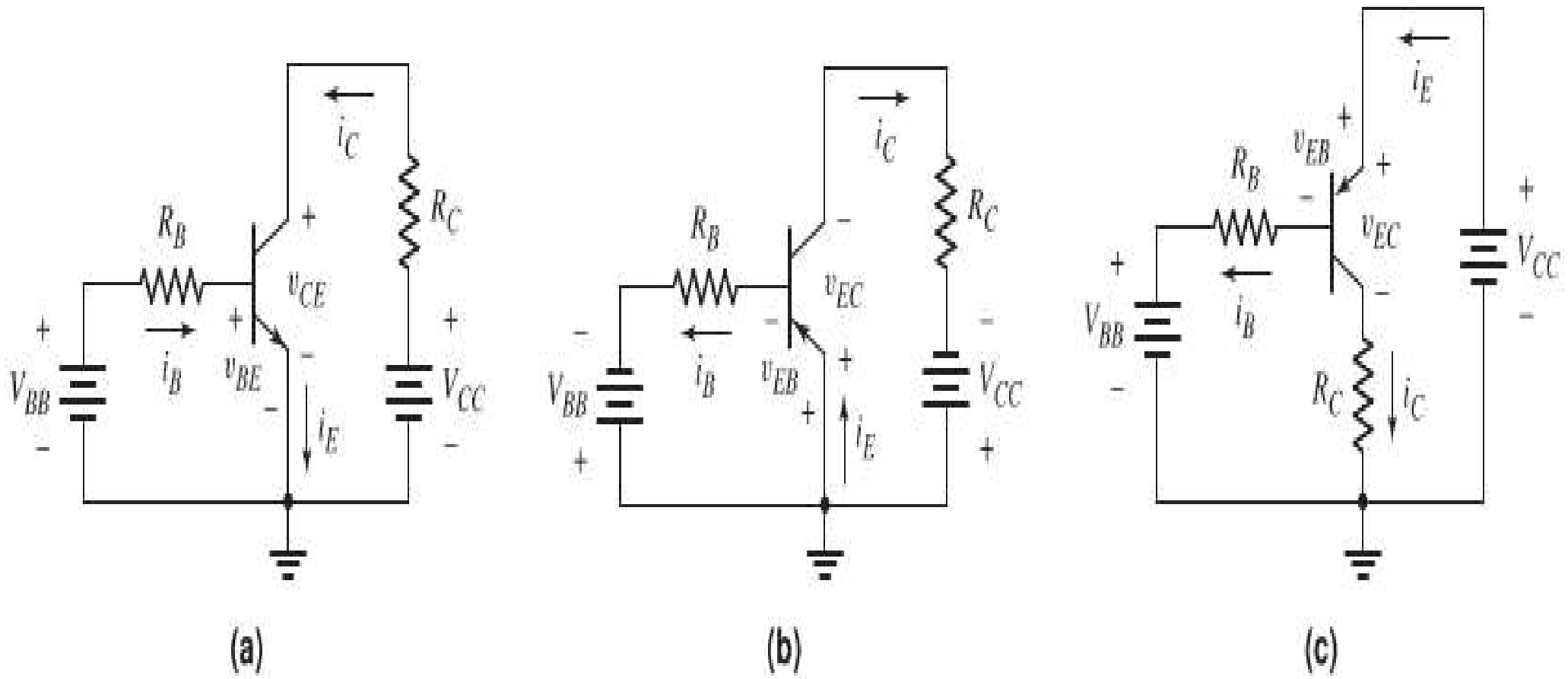


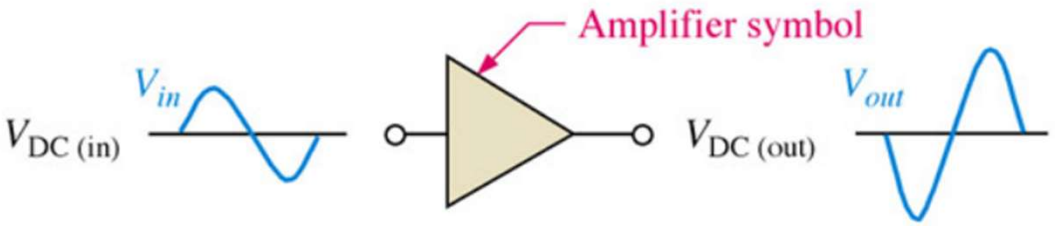
Figure 5.10 Common-emitter circuits: (a) with an npn transistor, (b) with a pnp transistor, and (c) with a pnp transistor biased with a positive voltage source

BJT Amplifier

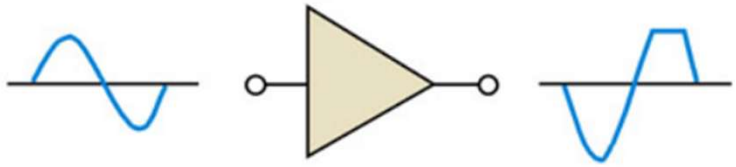
For the transistor to properly operate it must be biased. There are several methods to establish the DC operating point. We will discuss some of the methods used for biasing transistors as well as troubleshooting methods used for transistor bias circuits.

Amplification

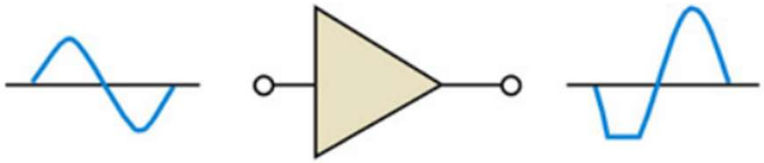
- The goal of amplification in most cases is to increase the amplitude of an ac signal without altering it.



(a) Linear operation: larger output has same shape as input except that it is inverted



(b) Nonlinear operation: output voltage limited (clipped) by cutoff

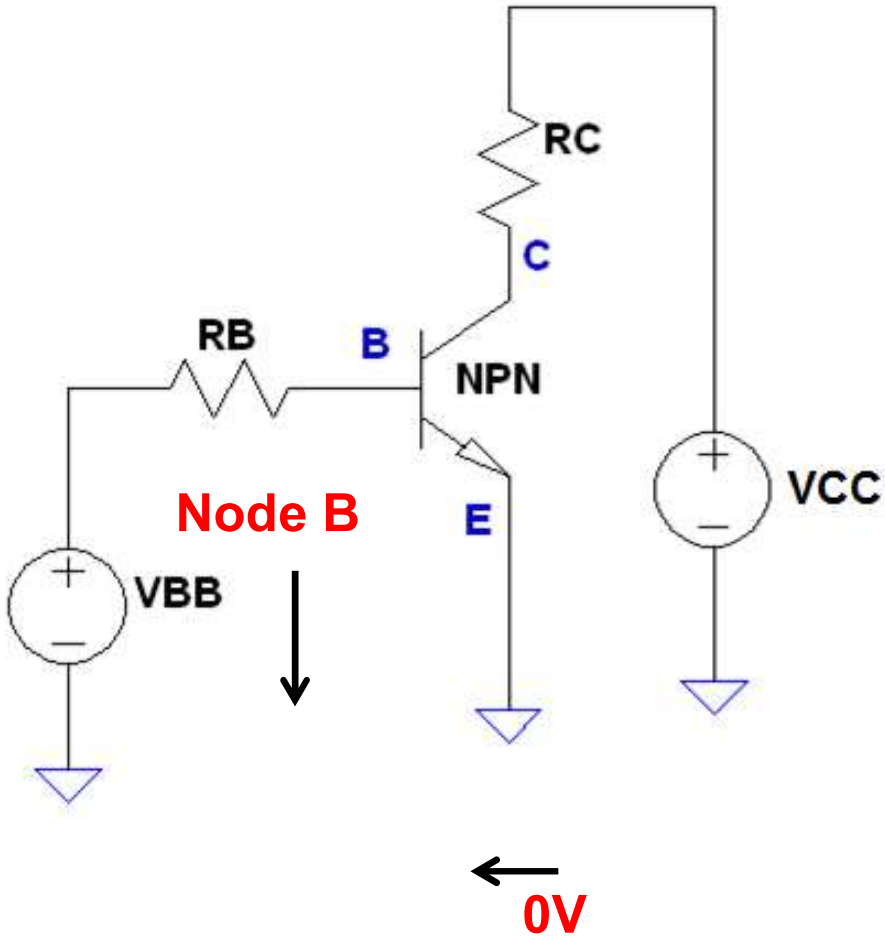


(c) Nonlinear operation: output voltage limited (clipped) by saturation

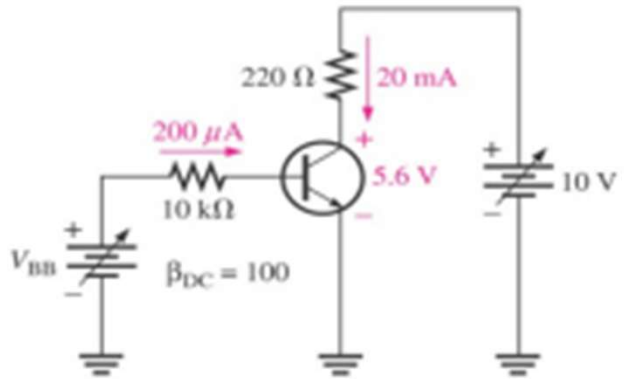
BJT: Current-Voltage Characteristic
I_C versus V_{CE}

Common-Emitter Configuration - npn

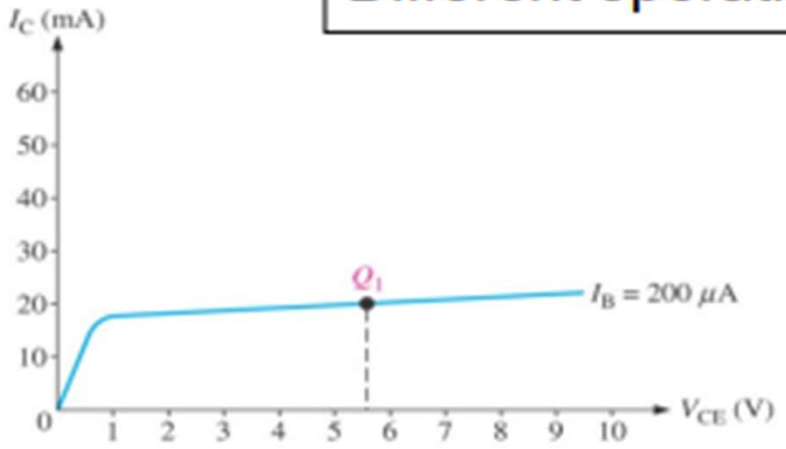
- The Emitter is common to both input (base-emitter) and output (collector-emitter).
- Since Emitter is grounded, $V_C = V_{CE}$
- With decreasing V_C (V_{CE}), the junction B-C will become forward biased too.
  - The current I_C quickly drops to zero because electrons are no longer collected by the collector

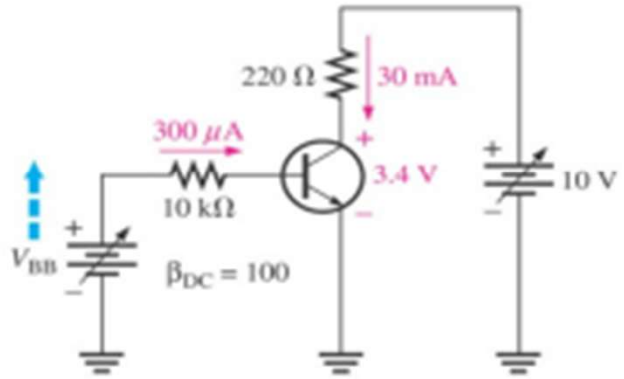


Different operating points

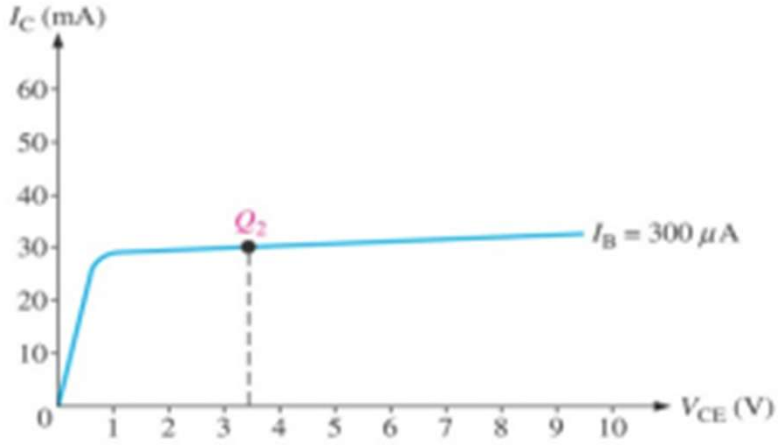


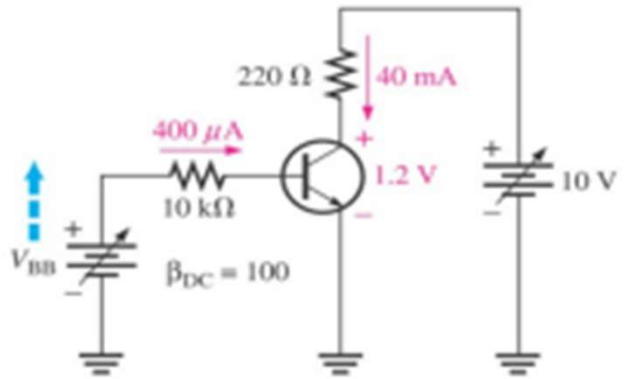
(a)



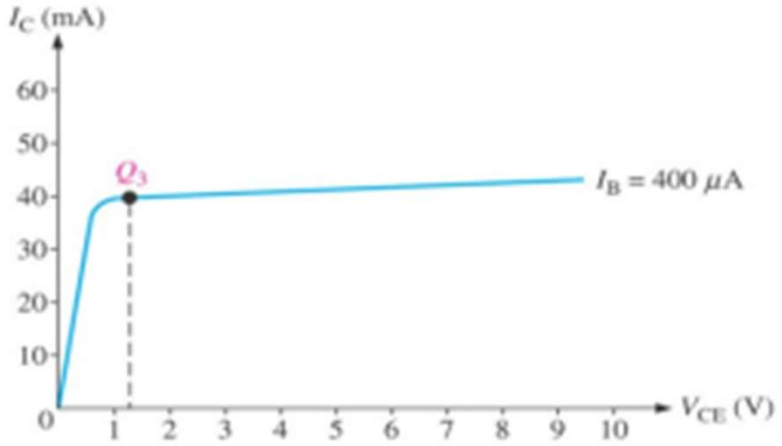


(b)





(c)



Q Point

- Assign three values to I_B and observe what happens to I_C and V_{CE}. First, V_{BB} is adjusted to produce an I_B of 200 μA as shown in Fig. Since $I_C = \beta_{DC} I_B$, the collector current is 20 mA as indicated, and V_{CE} as 5.6 V

$$V_{CE} = V_{CC} - I_C R_C$$

$$V_{CE} = 10V - (20mA)(220\Omega) = 10V - 4.4V = 5.6V$$

Q Point

V_{BB} is increased to produce an I_B of 300 µA and an I_C of 30 mA and V_{CE} as 3.4 V

$$V_{CE} = 10V - (30mA)(220\Omega) = 10V - 6.6V = 3.4V$$

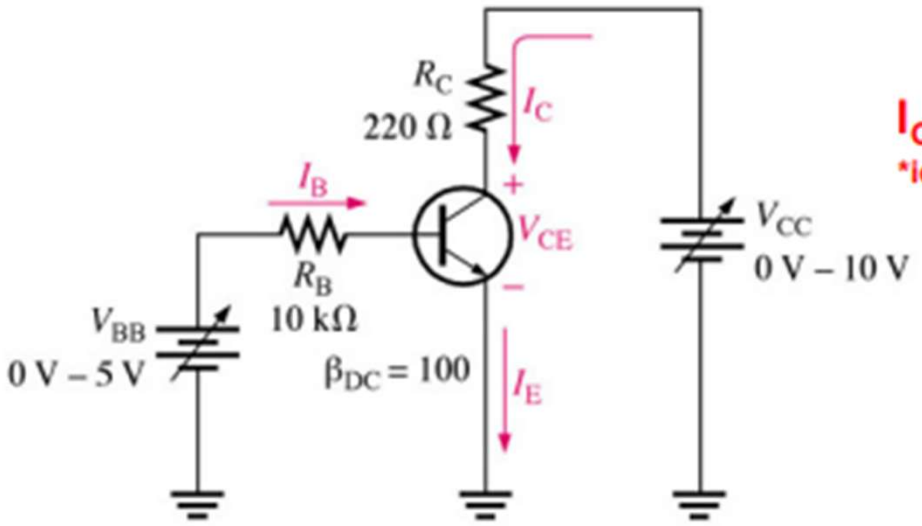
Finally, V_{BB} is increased to produce an I_B of 400 µA and an I_C of 40 mA and V_{CE} as 1.2 V

$$V_{CE} = 10V - (40mA)(220\Omega) = 10V - 8.8V = 1.2V$$

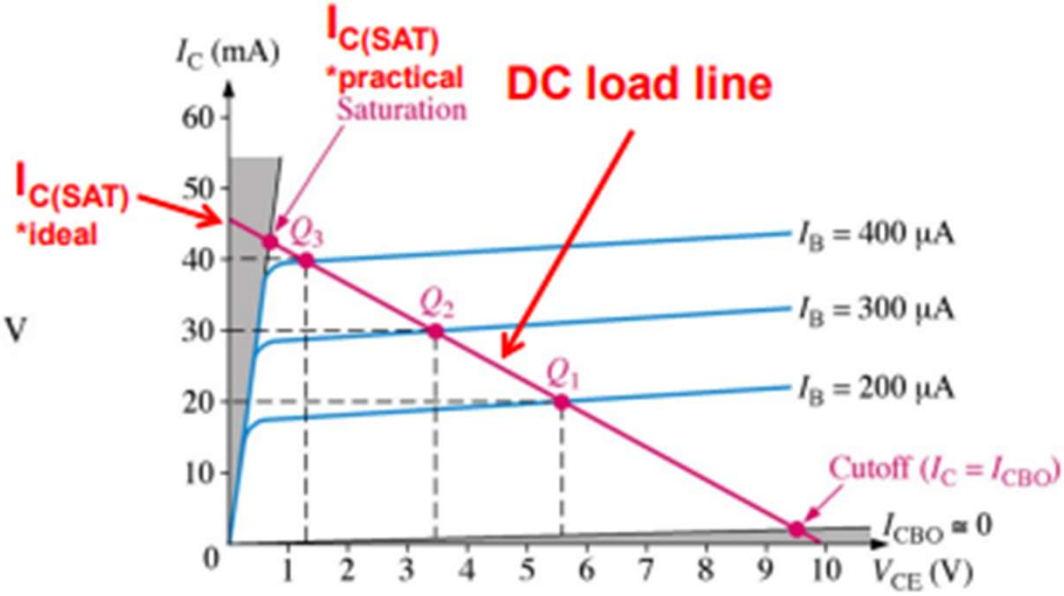
The DC Operating point

- For a transistor circuit to amplify it must be properly biased with dc voltages. The dc operating point is called the Q-point. The goal is to set the Q-point such that that it does not go into saturation or cutoff when an a ac signal is applied.

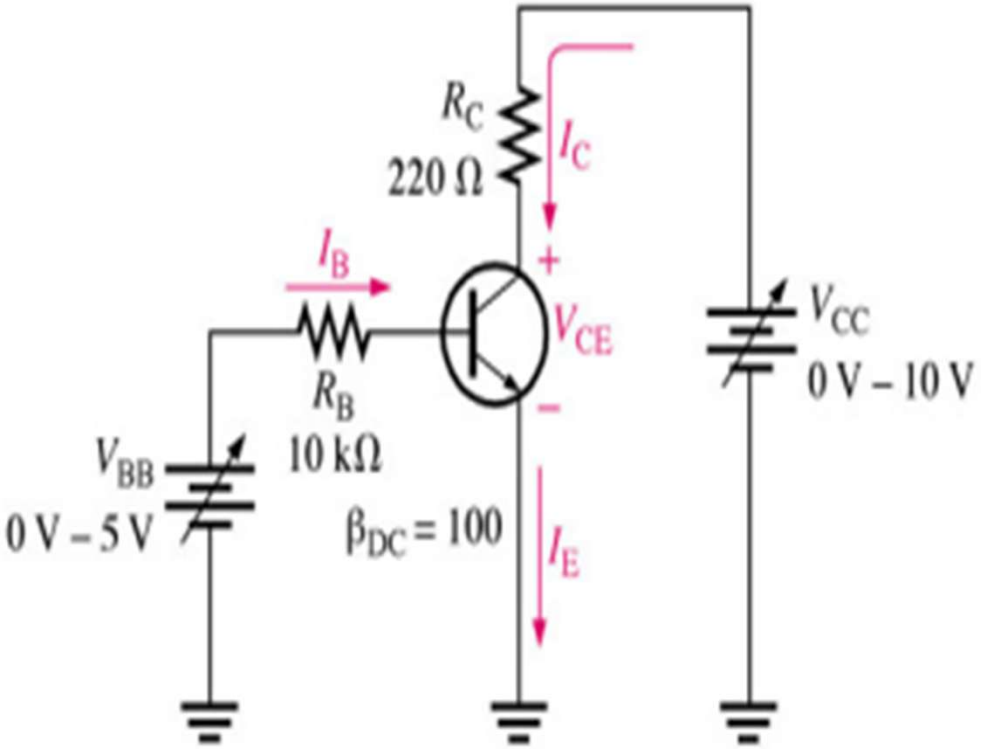
$$I_C = \frac{V_{CC} - V_{CE}}{R_C} = -\left(\frac{1}{R_C}\right)V_{CE} + \frac{V_{CC}}{R_C}$$



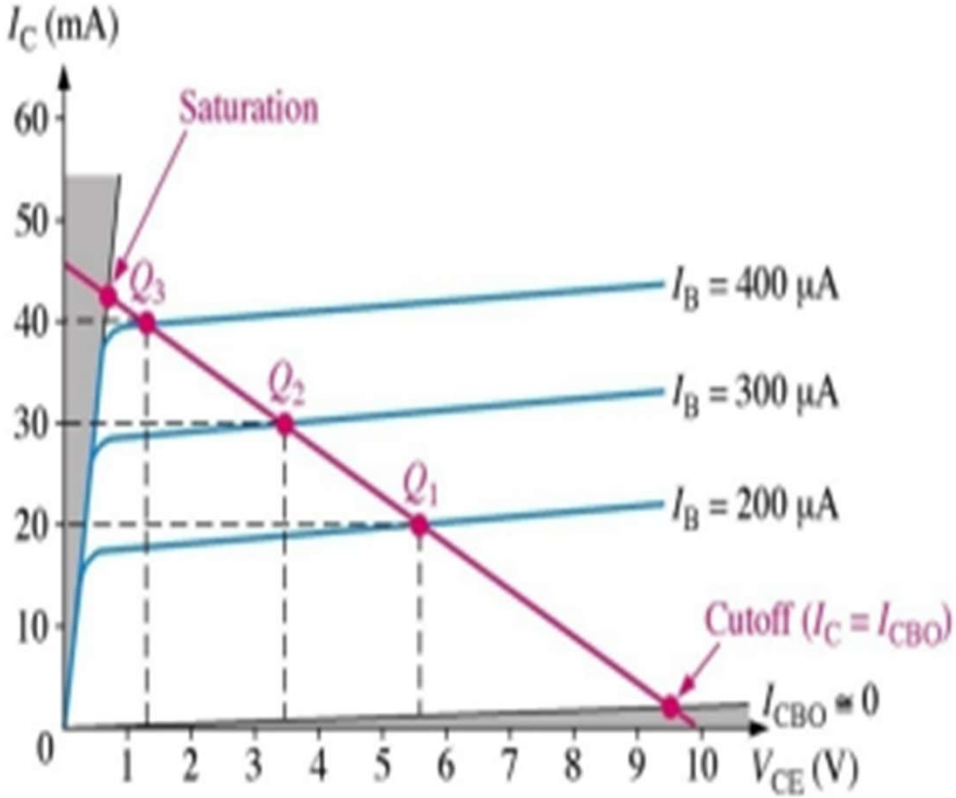
(a) DC biased circuit



- Recall that the collector characteristic curves graphically show the relationship of collector current and V_{CE} for different base currents. With the dc load line superimposed across the collector curves for this particular transistor we see that 30 mA of collector current is best for maximum amplification, giving equal amount above and below the Q-point. Note that this is three different scenarios of collector current being viewed simultaneously.



(a) DC biased circuit



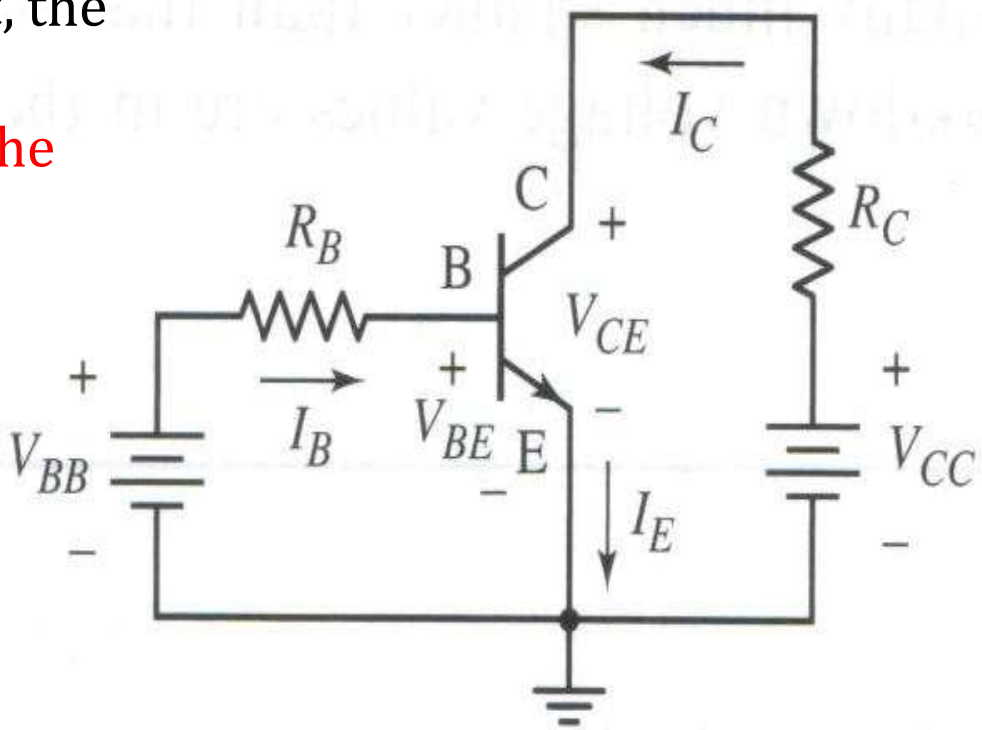
DC Load Line

- For the C-E portion of the circuit, the load line is found by writing Kirchhoff's voltage law around the C-E loop. We obtain:

$$V_{CE} = V_{CC} - I_C R_C$$

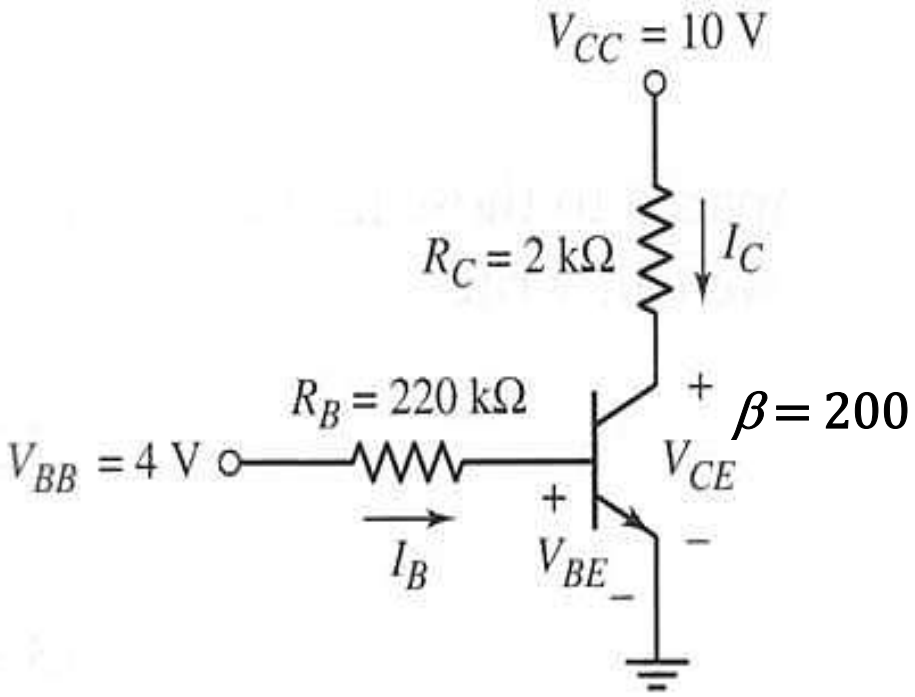
$$I_C = -\frac{V_{CE}}{R_C} + \frac{V_{CC}}{R_C}$$

$y = mx + c$



Write load line equation

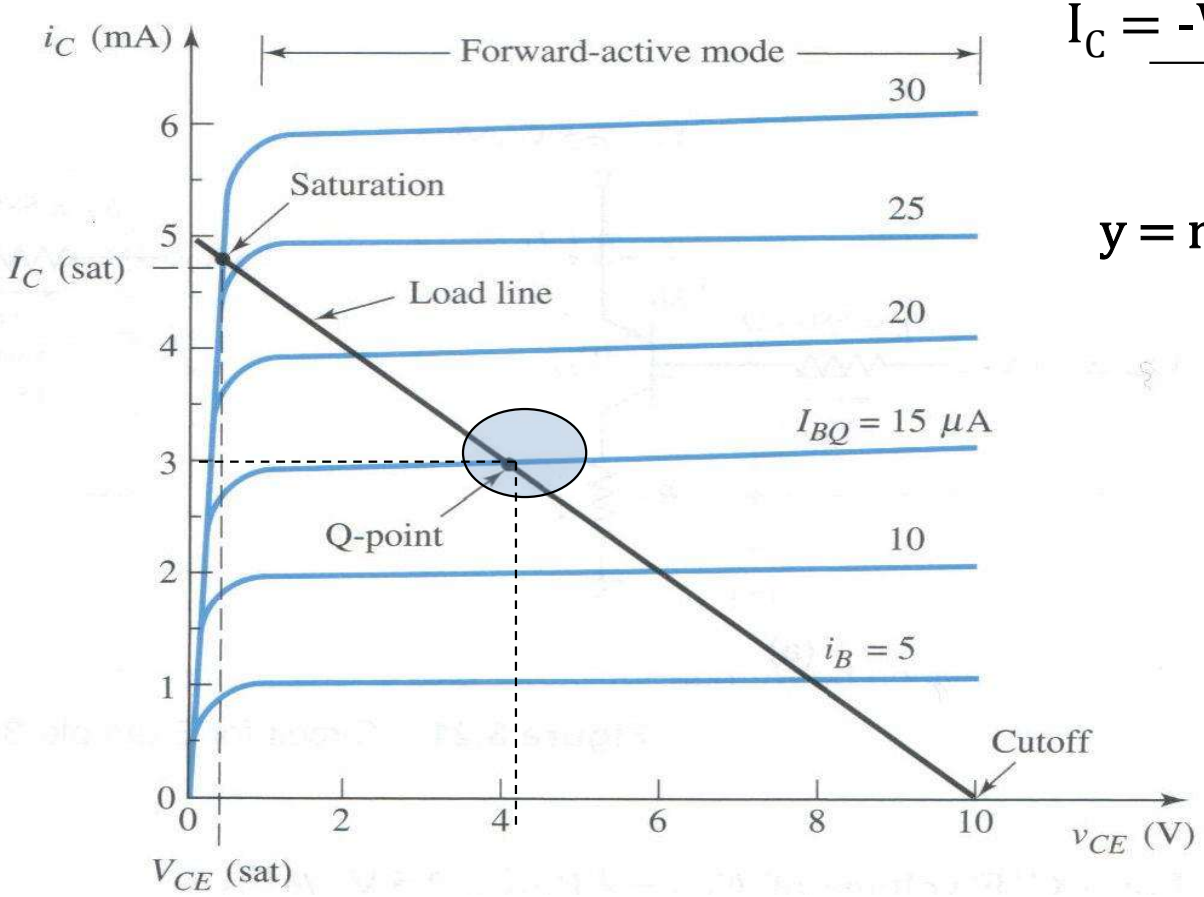
$$y = mx + c$$



$$V_{CE} = V_{CC} - I_C R_C$$

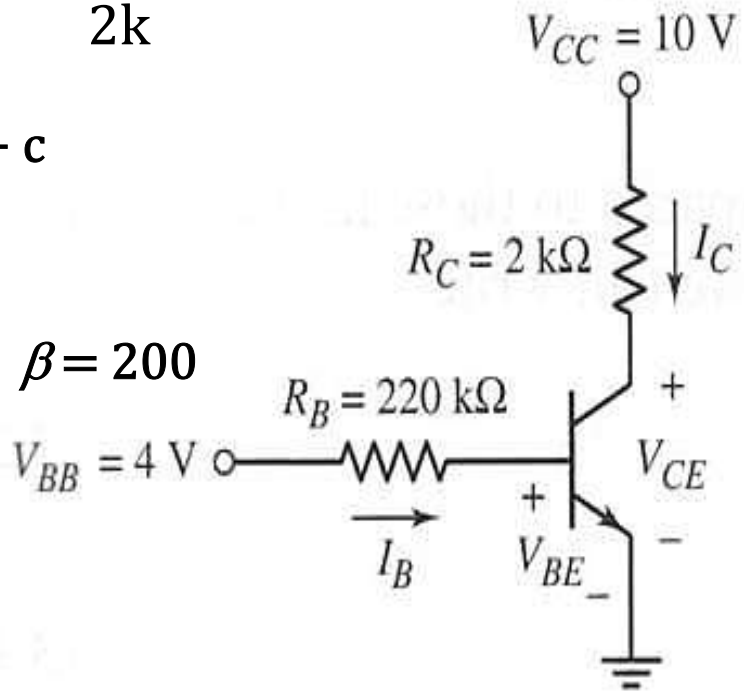
$$I_C = -\frac{V_{CE}}{R_C} + \frac{V_{CC}}{R_C}$$

Example



$$I_C = \frac{-V_{CE}}{2k} + \frac{10}{2k}$$

$$y = mx + c$$

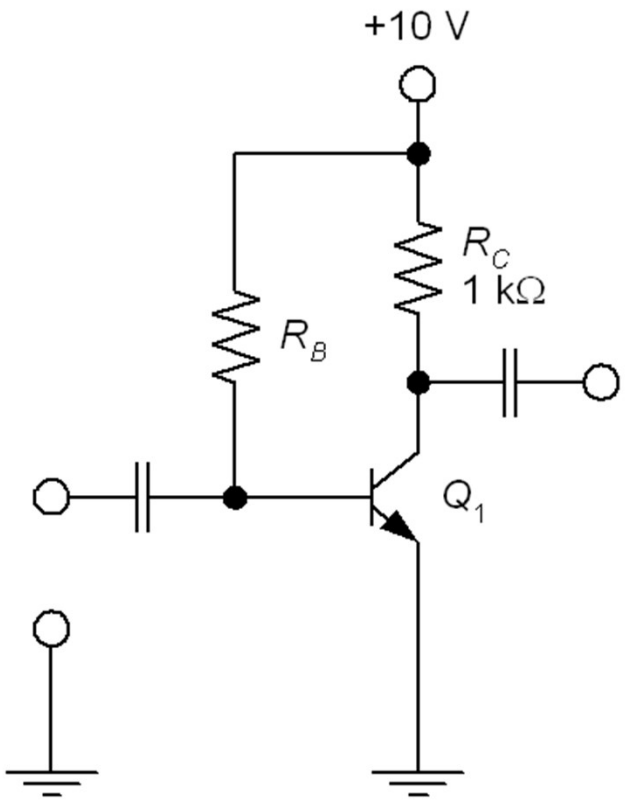


Q-point is the intersection of the load line with the i_C *vs* v_{CE} curve, corresponding to the appropriate base current

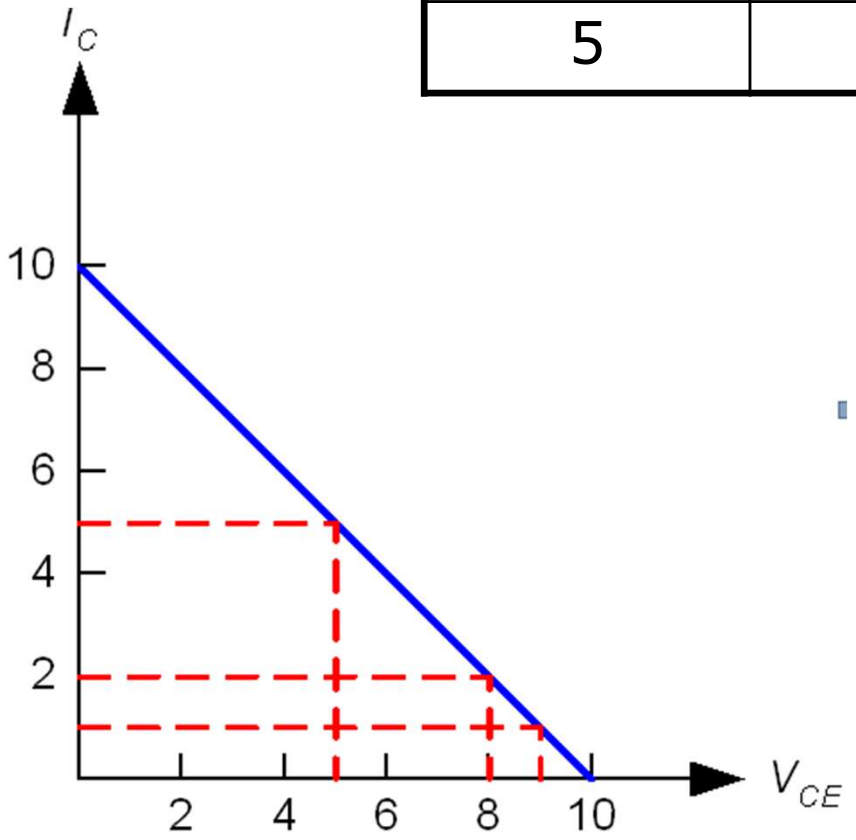
- To find the intersection points setting $I_C = 0$, $V_{CE} = V_{CC} = 10$ V
- setting $V_{CE} = 0$ $I_C = V_{CC}/R_C = 5$ mA

Plot the dc load line for the circuit shown in Fig. Then, find the values of V_{CE} for I_C = 1, 2, 5 mA respectively.

$$V_{CE} = V_{CC} - I_C R_C$$



I_C (mA)	V_{CE} (V)
1	9
2	8
5	5

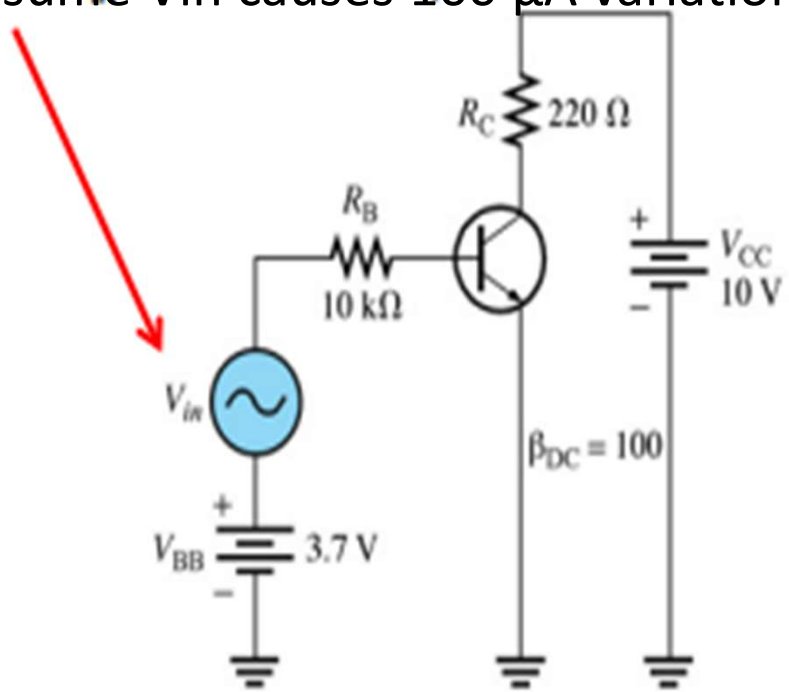


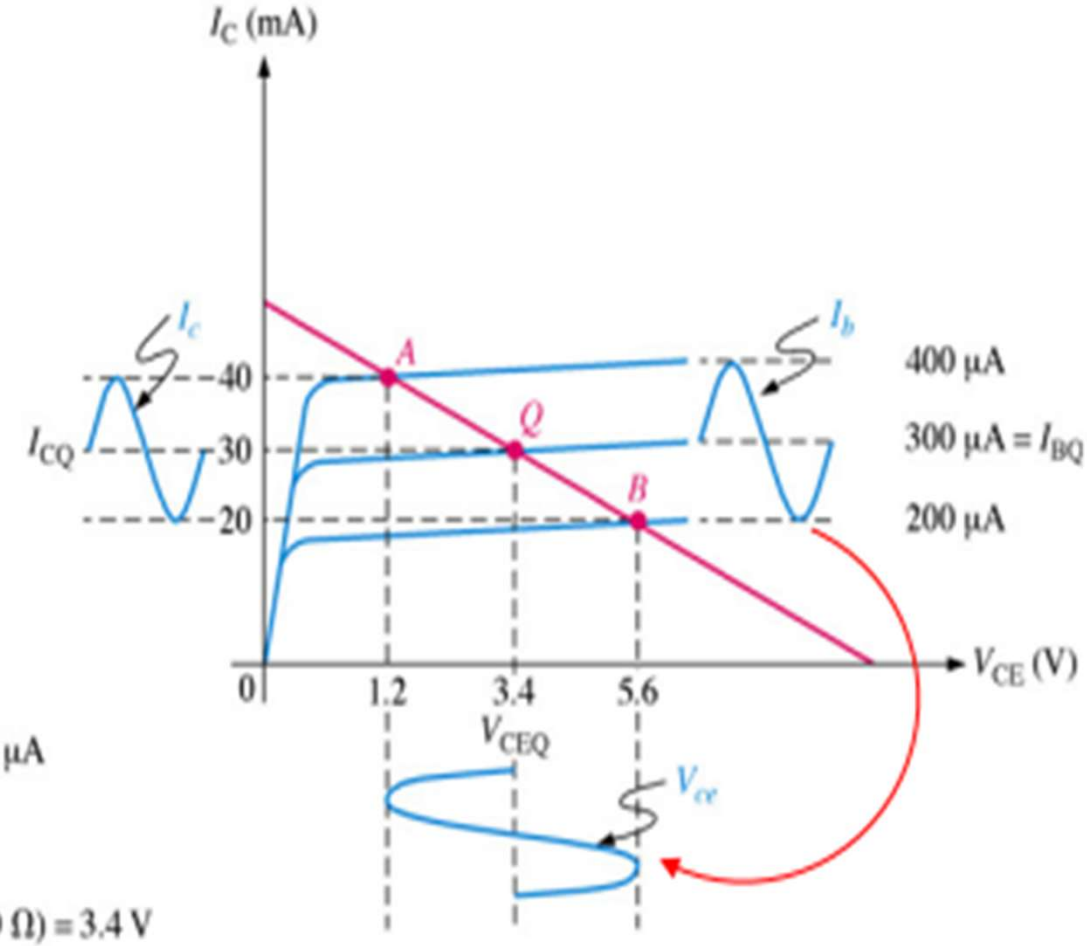
DC Load Line

- Describes graphically the DC operation of a transistor circuit
- A straight line drawn on the characteristic curves from the saturation value where $I_C = I_{C(sat)}$ on the y-axis to the
- Cutoff value where $V_{CE} = V_{CC}$ on the x-axis
- Determined by the external circuit (V_{CC} and R_C), not the transistor itself

- With a good Q-point established, let's look at the effect a superimposed ac voltage has on the circuit. Note the collector current swings do not exceed the limits of operation(saturation and cutoff). However, as you might already know, applying too much ac voltage to the base would result in driving the collector current into saturation or cutoff resulting in a distorted or clipped waveform.

Assume Vin causes 100 µA variation

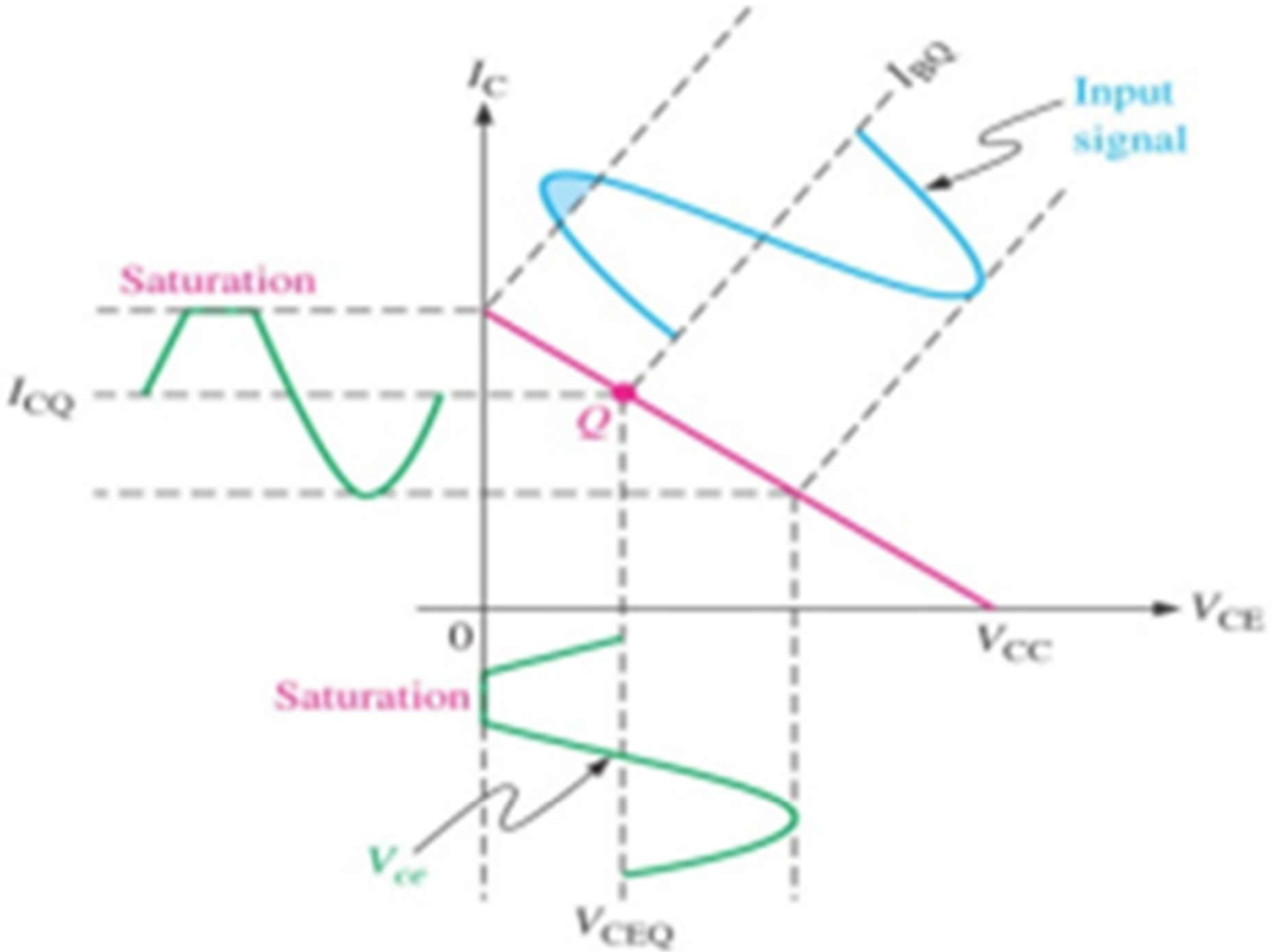




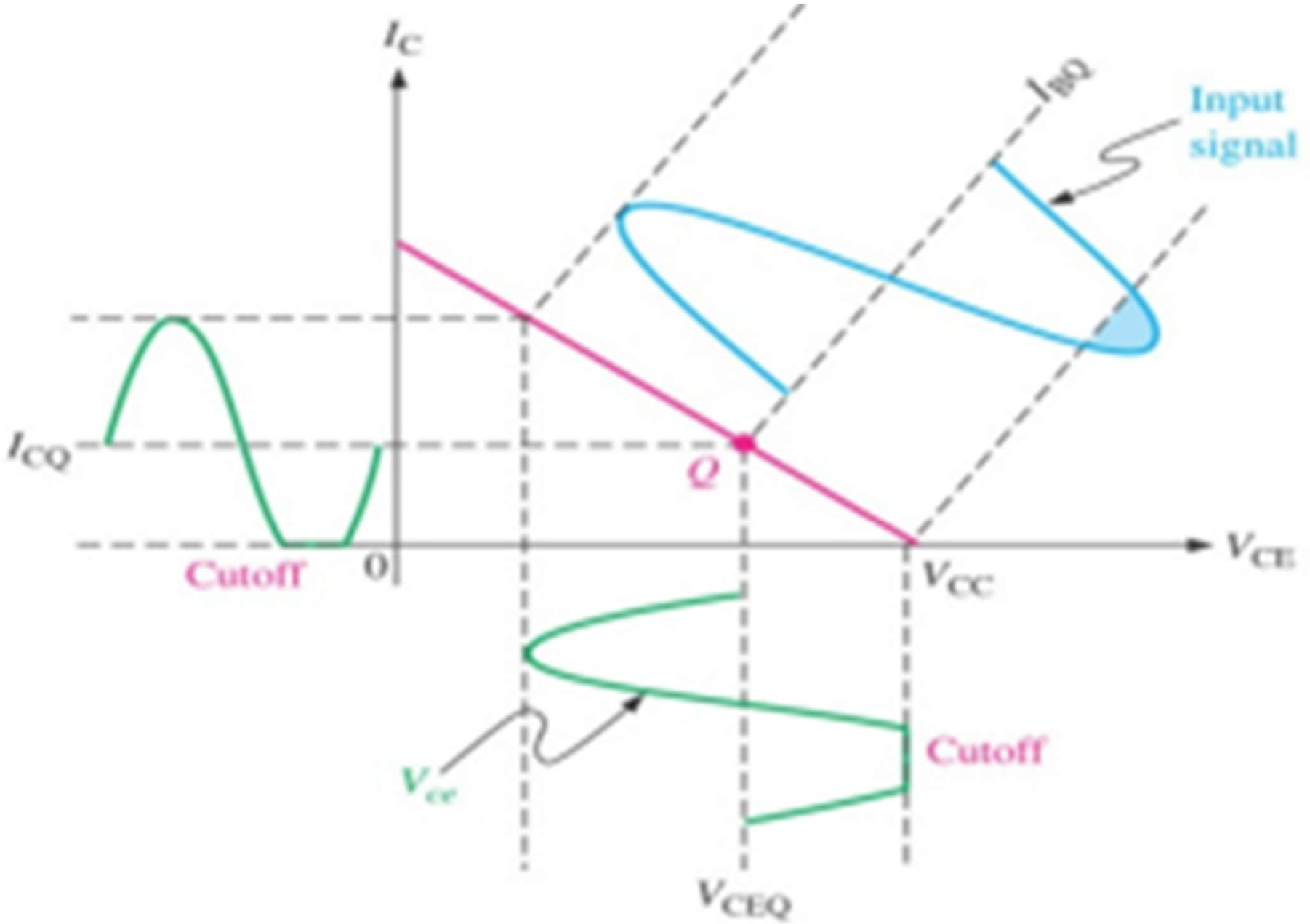
$$I_{BQ} = \frac{V_{BB} - 0.7\text{ V}}{R_B} = \frac{3.7\text{ V} - 0.7\text{ V}}{10\text{ k}\Omega} = 300\ \mu\text{A}$$

$$I_{CQ} = \beta_{DC} I_{BQ} = (100)(300\ \mu\text{A}) = 30\text{ mA}$$

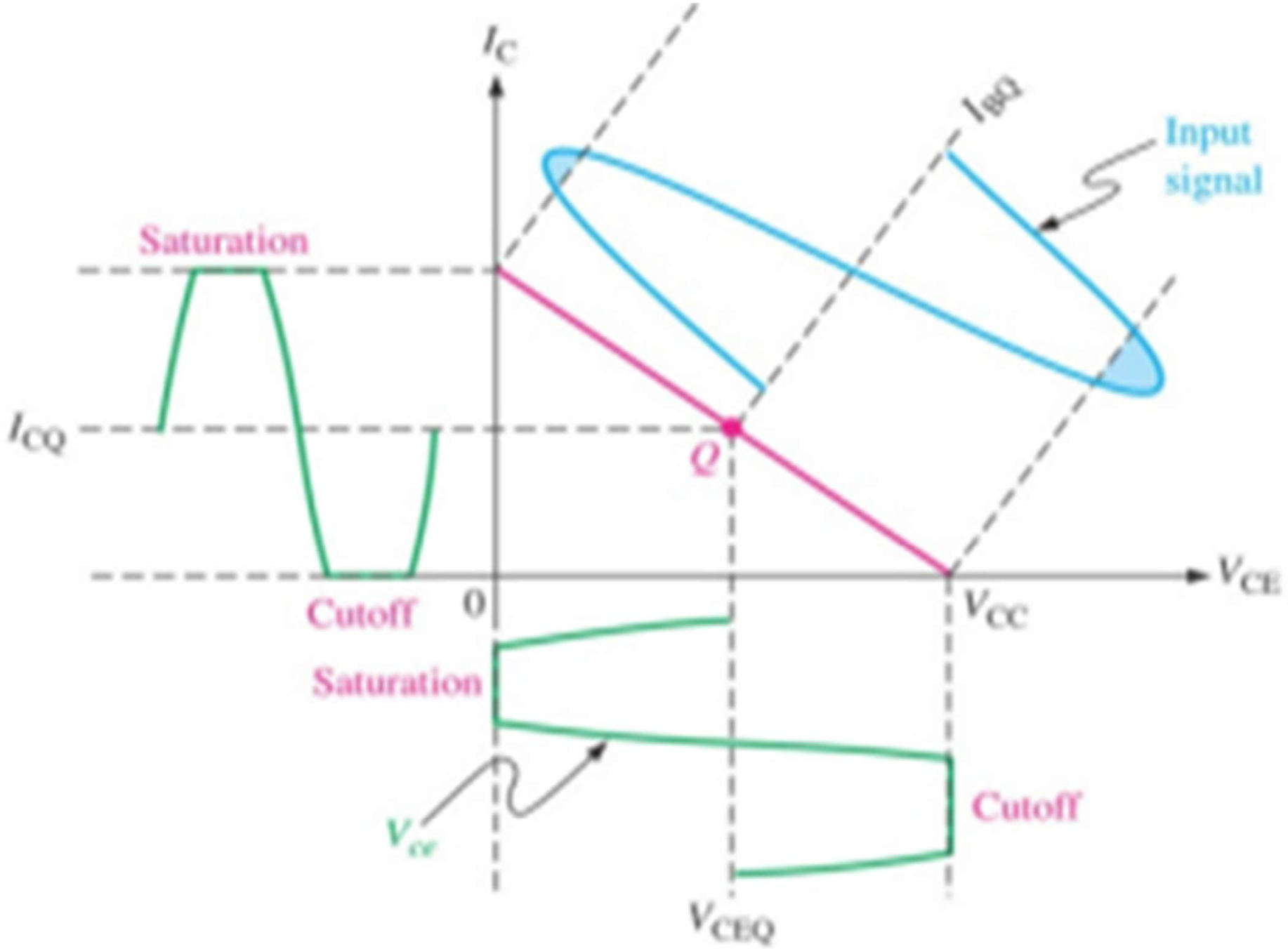
$$V_{CEQ} = V_{CC} - I_{CQ} R_C = 10\text{ V} - (30\text{ mA})(220\ \Omega) = 3.4\text{ V}$$



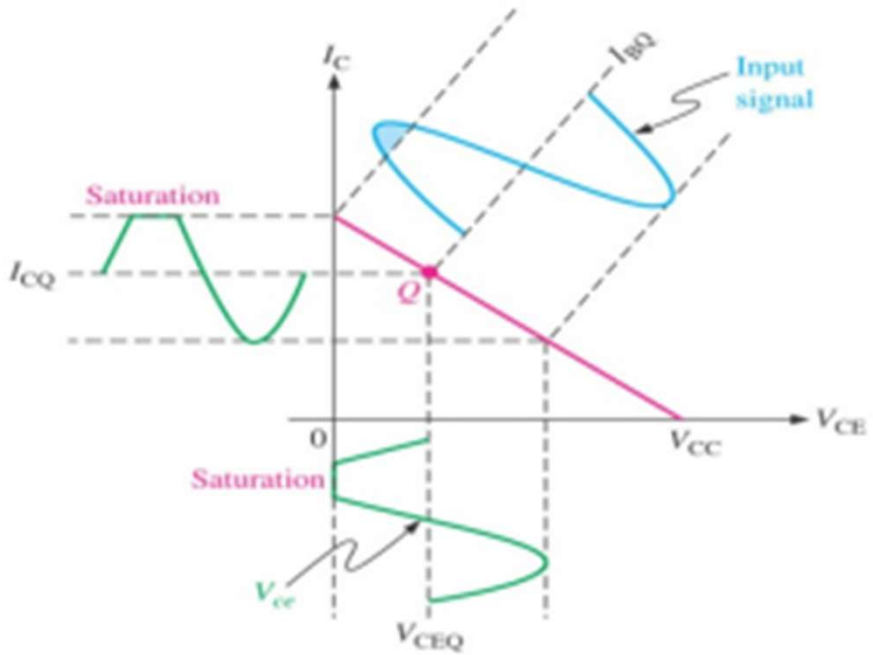
(a) Transistor is driven into saturation because the Q-point is too close to saturation for the given input signal.



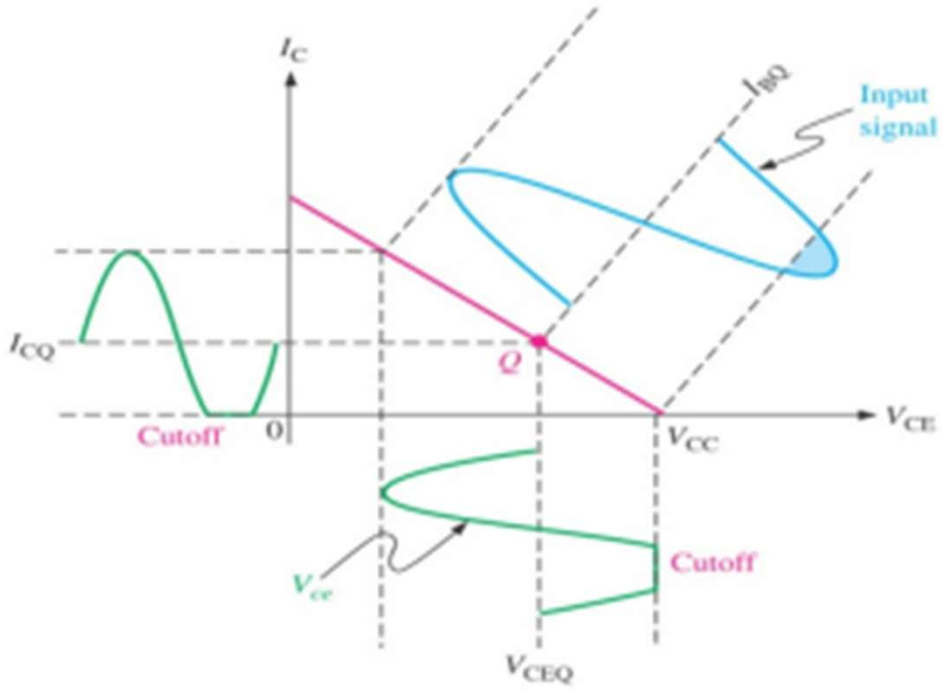
(b) Transistor is driven into cutoff because the Q-point is too close to cutoff for the given input signal.



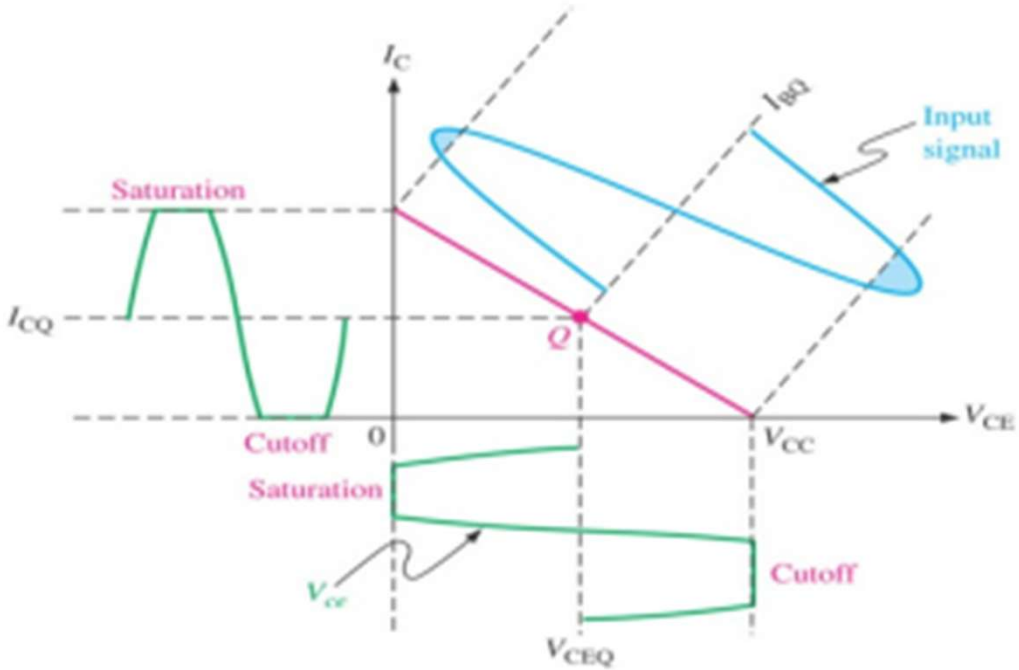
(c) Transistor is driven into both saturation and cutoff because the input signal is too large.



(a) Transistor is driven into saturation because the Q-point is too close to saturation for the given input signal.



(b) Transistor is driven into cutoff because the Q-point is too close to cutoff for the given input signal.



(c) Transistor is driven into both saturation and cutoff because the input signal is too large.

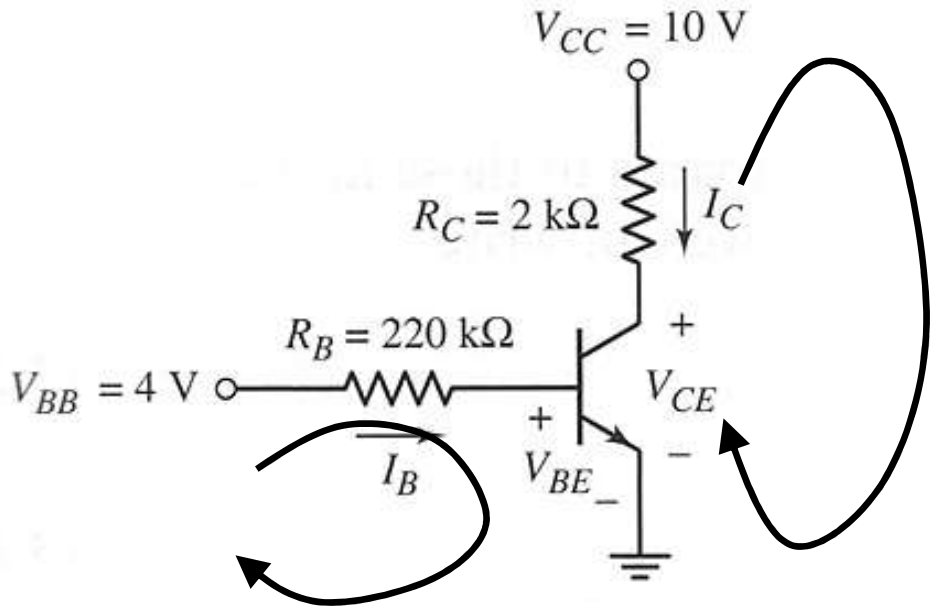
Examples

BJT DC Analysis

Common-Emitter Circuit

Example 1

Calculate the base, collector and emitter currents and the C-E voltage for a common-emitter circuit by considering $V_{BB} = 4$ V, $R_B = 220\text{k}\Omega$, $R_C = 2$ kΩ, $V_{CC} = 10$ V, $V_{BE}(\text{on}) = 0.7$ V and $\beta = 200$.



Solution: The base current is found as

$$I_B = \frac{V_{BB} - V_{BE}(\text{on})}{R_B} = \frac{4 - 0.7}{220} \Rightarrow 15\,\mu\text{A}$$

The collector current is

$$I_C = \beta I_B = (200)(15\,\mu\text{A}) \Rightarrow 3\,\text{mA}$$

and the emitter current is

$$I_E = (1 + \beta) \cdot I_B = (201)(15\,\mu\text{A}) \Rightarrow 3.02\,\text{mA}$$

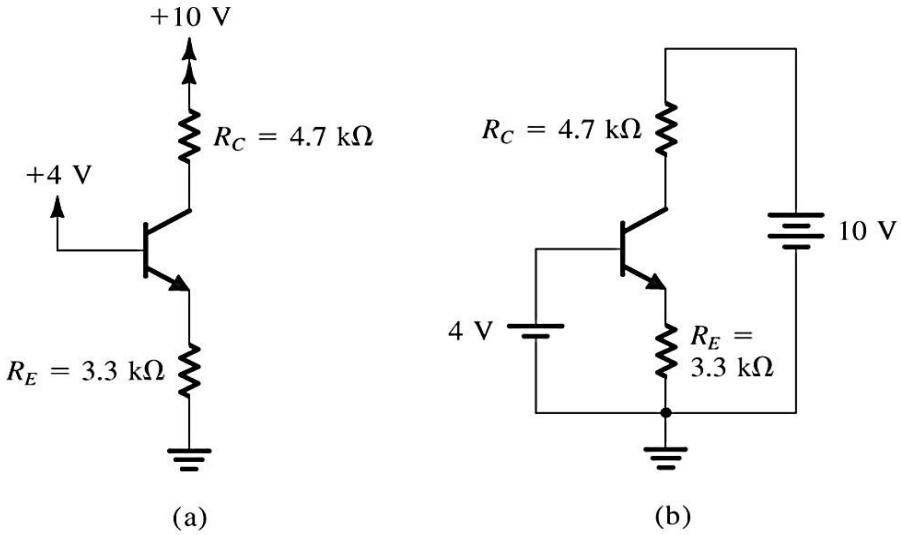
The collector-emitter voltage is

$$V_{CE} = V_{CC} - I_C R_C = 10 - (3)(2) = 4\,\text{V}$$

Example 2

Example 2

$\alpha = 0.99$



KVL at BE loop: $0.7 + I_E R_E - 4 = 0$

$I_E = 3.3 / 3.3 = \underline{\mathbf{1\ mA}}$

Hence, $I_C = \alpha\, I_E = \underline{\mathbf{0.99\ mA}}$

$I_B = I_E - I_C = \underline{\mathbf{0.01\ mA}}$

KVL at CE loop: $I_C R_C + V_{CE} + I_E R_E - 10 = 0$

$V_{CE} = 10 - 3.3 - 4.653 = \underline{\mathbf{2.047\ V}}$